Introduction to Theoretical Computer Science

Coursework 2

Due 12:00 on Friday 21.11.2025.

This coursework is summative and counts for 20% of the course grade. Submission via GradeScope, linked to from the LEARN page of the ITCS course, section Assessment.

Please remember the good scholarly practice requirements of the University regarding work for credit. You can find guidance at the School page `https://informatics.ed.ac.uk/taught-students/all-students/your-studies/academic-misconduct`. This also has links to the relevant University pages.

1 Computability

Let Σ be an alphabet and $A, B \subseteq \Sigma^*$ such that $A \cap B = \varnothing$.

Prove or disprove each of the following two claims.

Claim 1: If $\overline{A}$ and $\overline{B}$ are Turing-recognizable then there exists a decidable language C such that $A \subseteq C \ \wedge \ B \subseteq \overline{C}$. [*9 marks*]

Claim 2: If A and B are Turing-recognizable then there exists a decidable language C such that $A \subseteq C \ \wedge \ B \subseteq \overline{C}$. [*9 marks*]

2 Lambda-calculus

(a) Give a formal derivation for the type of the simply typed expression

$$\lambda f{:}\mathsf{nat} \to \mathsf{nat}.\lambda x{:}\mathsf{nat}.f(fx)$$

[*7 marks*]

(b) Reduce the following lambda term by *two* steps, showing each of the two steps. Also show any steps where you need to rename bound variables, if any.

Moreover, work out the most general type of this lambda term, or explain why the term cannot be typed.

$(\lambda x.\lambda y.xy)yx$ [*4 marks*]

(c) Reduce the following lambda term by *two* steps, showing each of the two steps. Also show any steps where you need to rename bound variables, if any.

Moreover, work out the most general type of this lambda term, or explain why the term cannot be typed.

$(\lambda x.\lambda y.y(xxy))(\lambda x.\lambda y.y(xxy))y$ [*4 marks*]

3 Complexity

The h-SCHEDULE problem is defined as follows. You are given a list of exams $X = \{x_1, \ldots, x_m\}$ to be scheduled, and a list of students $S = \{s_1, \ldots, s_n\}$. Each exam is taken by some specified subset of the students. You must schedule these exams into slots so that no student is required to take two exams in the same slot. The problem is to determine whether there exists such a schedule that uses just h slots.

(a) Formulate the h-SCHEDULE problem precisely, e.g., by introducing suitable functions to specify the relation between exams, students and slots. [*3 marks*]

(b) Show that h-SCHEDULE is in NP. [*2 marks*]

(c) The k-COLOUR problem is as follows. You are given a graph $G = \langle V, E\rangle$. You must find an assignment of colours to vertices such that each edge connects vertices of different colours.

Formulate the k-COLOUR problem precisely, e.g., by introducing a suitable function to specify how vertices are coloured. [*3 marks*]

(d) Show that k-COLOUR is in NP. [*2 marks*]

(e) Give a polynomial time reduction from h-SCHEDULING to k-COLOUR. [*7 marks*]